

Case Study: The Shanahan Tree

Wide zone, bootlegs, a fullback and yards after the catch: how one family of coaches built an offense that looks like one play and works with almost any quarterback

Sunday, first-and-10 at midfield. San Francisco lines up with a fullback, a tight end and two receivers squeezed in tight to the formation, the quarterback under center. A receiver jogs in motion across the formation, the ball is snapped, and all five offensive linemen take the same hard step to their right. The running back takes the handoff on a path toward the tight end, the linebackers sprint sideways to meet him, and he cuts upfield for six yards. Two plays later: the same personnel, the same formation, the same motion, the same step by the line, the same back running the same path. Except this time the quarterback still has the ball. He is rolling the other way, alone, and the tight end who started on the right side of the formation is running free across the field on the left. He catches it in stride and runs for 20 more yards after the catch. The broadcast calls it “a busted coverage.” It wasn’t. Both plays were built to look identical for the first second, and that one second is the whole idea. This chapter is about the family of coaches who built an offense around it, and about why their offense seems to work no matter who plays quarterback.

You’ll be able to:

- Trace the outside-zone (wide-zone) system from Alex Gibbs’s and Mike Shanahan’s 1990s Broncos to Kyle Shanahan’s 49ers, and name the problem it was built to solve at each step.
- Explain how wide zone, the fullback, pre-snap motion and yards-after-catch passing fit together as one idea, and say what each piece takes away from a defense.
- Diagram wide zone against a bear front and the bootleg-keeper that punishes the defender the offense leaves unblocked, and call which one is coming from how that defender is playing.
- Explain why the system makes quarterbacks efficient, and weigh both sides of the “system quarterback” debate with EPA per dropback.
- Read the system’s data fingerprint: under-center rate, play-action rate, 21 personnel, YAC over expected and EPA per play.
- Name the tree’s play-callers in the 2025 season and in the 2026 season, and keep the two lists apart.

You’ll need

- **The Modern Era:** the 2010s and 2020s in one arc; spread passing went league-wide, defenses answered with lighter personnel and then two-high safeties, and offenses answered back with heavier personnel, motion and a return to the quarterback lining up under center.
- **Constraint Theory and Sequencing:** a *base play* the offense wants to run all day, *constraint plays* that punish a defense for cheating against it, the constraint tree around outside zone, and the illusion of complexity (a few plays dressed to look like many, many plays dressed to look like one).
- Also used: outside zone (the line steps sideways together and the back makes one cut), the bootleg, naked bootleg and keeper, YAC (yards after the catch), and EPA (expected points added: how much a play changed the offense’s expected score; above zero is good for the offense).

How to read the diagrams. Offense is blue circles, defense orange squares. *Left* and *right* always mean the offense’s left and right. LT, LG, C, RG and RT are the five offensive linemen, Q the quarterback, R the running back and F the fullback (a second back, lined up closer to the line, whose main job is blocking). Y is the tight end, X and Z the wide receivers. On defense, linemen are labelled by technique number (a 0 is head-up on the center, a 3 on a guard’s outside shoulder, a 9 outside the tight end; E is an end), M and W are the Mike and Will linebackers, SS and FS the safeties and CB the cornerbacks. Lines ending in a short bar are blocks, arrows are paths, and the thick dark arrow is the ball carrier.

Two problems, thirty years apart

Every successful offense is an answer to a question. The Shanahan family’s offense was built twice, by a father and then by his son, and each time it answered a different question.

Denver, 1995: the defense won't stand still

When **Mike Shanahan** became the Denver Broncos' head coach in 1995 he brought two things together. The first was the passing game of the San Francisco 49ers: he had just spent three seasons (1992–94) as George Seifert's offensive coordinator, running the West Coast offense with Steve Young and winning Super Bowl XXIX.¹ The second was a line coach, **Alex Gibbs**, who had a theory about running the ball.

The problem in front of them was the defenses of the mid-1990s. Dom Capers's Pittsburgh *zone blitzes* sent linebackers into gaps and dropped linemen into coverage, and fronts slanted and stunted on almost every snap. A line that blocked "the man in front of me" kept finding that the man had left and somebody else had arrived. Gary Kubiak, Denver's offensive coordinator under Shanahan, later remembered that in a man scheme "you're getting picked off and you look bad."² Gibbs's answer was outside zone, also called wide zone: every lineman steps the same way at the snap and blocks whoever enters his area, so a defense that moves is still blocked, and the running back reads the blocks and cuts once. Zone Running builds the play from the ground up; here what matters is the *philosophy* Gibbs attached to it. He wanted very few runs, repeated until the linemen could block them against any front, lighter and quicker linemen who could run sideways, and backs who would press the hole and make one decisive cut. Gibbs was famous among coaches for his blunt teaching and his near-religious faith in repetition; his clinic tapes are still sold to high school coaches.³

Shanahan supplied the other half: the **bootleg**, the pass that looks exactly like the run until the quarterback turns the other way. John Elway, in the last four seasons of his career, faked wide zone and rolled away from it again and again. Denver won Super Bowls after the 1997 and 1998 seasons; Terrell Davis, a sixth-round pick, ran for 2,008 yards in 1998 and was named MVP, and when Davis got hurt a string of backs who were not first-round picks kept producing 1,000-yard seasons in the same scheme.⁴ The lesson the league took was that the system, not the star, was doing a lot of the work. Hold on to that idea; it comes back, aimed at quarterbacks, later in this chapter.

San Francisco, 2017: no quarterback, and defenses built for the spread

Twenty-two years later Mike's son **Kyle Shanahan** took over a 49ers team that had gone 2–14 in 2016.⁵ Kyle had grown up in the Denver building, played receiver at Texas, and coached under Gary Kubiak in Houston, under his father in Washington, and in Cleveland and Atlanta, where as offensive coordinator in 2016 his offense led the league in scoring and Matt Ryan was named MVP.⁶ He inherited a roster with no starting quarterback (his first starters were Brian Hoyer and the rookie C. J. Beathard) and few skill players a defense feared.

His problem was not the zone blitz. It was the defenses of the 2010s. To stop spread passing, teams had made the nickel (five defensive backs) their real base defense and filled the field with fast, light linebackers and safeties who could run with receivers. How do you beat a defense like that when you can't count on your quarterback to win from the pocket on third-and-8?

Kyle's answer kept his father's run-and-boot core and added two things on top. First, make the defense *declare* its personnel and its coverage before the snap, by using personnel it doesn't like (a fullback) and motion it has to react to. Second, design the passing game so that the hard part of a completion is done by the play, not the passer: throws to receivers already running, in space that the run fake has emptied, caught by players chosen because they can run after the catch. A defense that is fast and light is exactly the defense that has to run sideways to stop wide zone, and exactly the defense that has left grass behind its linebackers when the ball comes out on a bootleg.

¹Mike Shanahan was the 49ers' offensive coordinator 1992–94 (Super Bowl XXIX after the 1994 season) and Denver's head coach 1995–2008: Pro Football Reference, <https://www.pro-football-reference.com/coaches/ShanMi0.htm>.

²Kevin Seifert, "The secret to the NFL's best rushing teams: Why zone blocking is here to stay," ESPN, Dec 23, 2019 (Kubiak on the zone blitzes Denver faced): https://www.espn.com/nfl/story/_/id/28353469/the-secret-nfl-best-rushing-teams-why-zone-blocking-here-stay.

³Gibbs coached Denver's line 1984–87 and again 1995–2003, and died July 12, 2021, aged 80: ProFootballTalk, "Alex Gibbs dies at 80" (Jul 12, 2021): <https://www.nbcsports.com/nfl/profootballtalk/rumor-mill/news/alex-gibbs-dies-at-80>. On his philosophy (two runs, technique over size on the line, one-cut backs): Chris B. Brown, Grantland, Jan 10, 2012: <https://grantland.com/features/chris-brown-gary-kubiak-houston-texans-running-game/>. Clinic video: "Alex Gibbs: Outside Zone, Tight Zone, Boots and Thoughts on the Power" (CoachTube): <https://coachtube.com/course/football/alex-gibbs-outside-zone-tight-zone-boots-and-thoughts-on-the-power/14602600>.

⁴Pro Football Hall of Fame, Terrell Davis (sixth round, 1995; 2,008 rushing yards and MVP in 1998): <https://www.profootballhof.com/players/terrell-davis/>. Denver's later 1,000-yard backs under Shanahan (Olandis Gary, Mike Anderson, Clinton Portis, Reuben Droughns, Tatum Bell): Pro Football Reference team pages, <https://www.pro-football-reference.com/teams/den/>; the list is checked in this book's Zone Running notes.

⁵2016 San Francisco 49ers, 2–14: <https://www.pro-football-reference.com/teams/sfo/2016.htm>. 2017 starters (Hoyer, Beathard, Garoppolo, who arrived in a trade with New England in October 2017): <https://www.pro-football-reference.com/teams/sfo/2017.htm>.

⁶Kyle Shanahan's coaching career (Buccaneers 2004–05; Texans 2006–09, offensive coordinator 2008–09; Washington offensive coordinator 2010–13; Browns 2014; Falcons 2015–16; 49ers head coach from 2017): Pro Football Reference, <https://www.pro-football-reference.com/coaches/ShanKy0.htm>. The 2016 Falcons led the NFL with 540 points and Matt Ryan was the AP MVP: <https://www.pro-football-reference.com/teams/atl/2016.htm>. Titles by year (Tampa Bay offensive quality control; Houston receivers 2006, quarterbacks 2007, coordinator from January 2008): Texas Athletics, https://texaslonghorns.com/news/2010/1/22/012210aa_409.

! Why it exists

The Shanahan offense exists to make the defense's speed and movement work for the offense. Zone blocking means slants and blitzes don't leave anyone unblocked; the run's sideways flow pulls fast defenders out of position; the bootleg and play-action passes look identical to the run for the first second and throw into the space the flow empties. **The trade:** it needs a particular roster (quick linemen, one-cut backs, a fullback, receivers who block and run after the catch), it asks for a lot of patience on early downs, and it is at its least natural when there is nothing to fake: third-and-long, two-minute drills, playing from far behind. Whether that last weakness is real is one of the things the data will test.

The answer: four ingredients that are one idea

Most offenses can be described as a list of plays. The Shanahan offense is better described as one picture shown over and over, with four ingredients mixed in different amounts. Learn the four and why each one is there, and you can recognise the family from the stands.

1. Wide zone, the play they will run all day

Wide zone is the *base play*, the play the whole offense is organised to protect. From Constraint Theory, a base play is the one you want to be good at no matter what the defense does, and the plays around it exist to stop the defense from cheating against it. Shanahan teams treat wide zone exactly that way. They will run it into a defense that is stopping it, on purpose, because every run *teaches* the defense something: the linebackers learn to flow hard with the line's first step, the backside end learns to chase the back from behind, and the safety learns to come down fast. Every lesson the defense learns is a habit that a later play will punish.

That patience produces a kind of running back the scheme seems to manufacture. Denver's string of late-round 1,000-yard backs had a modern sequel in San Francisco: Raheem Mostert, an undrafted free agent who ran for 220 yards in the 2019 NFC Championship Game, and Elijah Mitchell, a sixth-round rookie who ran for 963 yards in 11 games in 2021.⁷ (The reverse is also true: when the 49ers had an elite back, Christian McCaffrey, he was the league's Offensive Player of the Year in 2023.) The scheme doesn't make every back great, but it makes the job simple enough that an ordinary one can succeed: press the aiming point, see one crease, cut once.

2. The fullback, a personnel problem for the defense

In the 2010s, the fullback nearly vanished from the NFL; many teams carried none. Shanahan's 49ers made him a centrepiece, signing Kyle Juszyk in 2017 to what was then the richest contract ever given to a fullback.⁸ Why spend money on a position the rest of the league had abandoned?

Because a fullback puts the defense in a bind before the snap. Two backs and one tight end (21 personnel) is a running look, and the defense has two bad choices:

- **Stay in nickel** (five defensive backs, two linebackers) and the offense has an extra blocker at the point of attack: the fullback leads into the hole on wide zone, and a light box gets run over.
- **Go to base defense** (a third linebacker for the fifth defensive back) and the defense has put a slower player on the field against an offense whose fullback, tight end and back can all catch the ball. Now the play-action pass is a linebacker trying to cover George Kittle.

The fullback is also the offense's most flexible piece once the ball is snapped. He can lead on wide zone, kick out an end on split zone, block the backside end on a protected bootleg, or *slice* across the formation into the flat as the bootleg's short receiver, all from the same spot behind the quarterback. Teams that copy the system without a fullback use a tight end or H-back for the same jobs.

The numbers show how unusual that is. From 2019 to 2025 the 49ers lined up in 21 personnel on 34% of their plays, the most in the league over the period, and they ranked first or second in every one of those seasons; the median team was in 21 personnel on

⁷Mostert, 2019 NFC Championship (Jan 19, 2020): 29 carries, 220 yards, 4 TD: <https://www.pro-football-reference.com/boxscores/202001190sfo.htm>. Elijah Mitchell, sixth-round pick in 2021, 963 rushing yards in 11 games (nflverse player stats, 2021 regular season; checks with <https://www.pro-football-reference.com/players/M/MitcEl00.htm>). McCaffrey, AP Offensive Player of the Year 2023: <https://www.nfl.com/news/niners-rb-christian-mccaffrey-named-ap-2023-nfl-offensive-player-of-the-year>.

⁸Juszyk agreed to a four-year, \$21 million contract with San Francisco on March 8, 2017 (\$5.25 million a year), reported at the time as the largest ever for a fullback: CBS Sports, "Here's why fullbacks were the biggest winners in the NFL on Wednesday," <https://www.cbssports.com/nfl/news/heres-why-fullbacks-were-the-biggest-winners-in-the-nfl-on-wednesday>. He has since topped it himself (five years, up to \$27 million, in 2021).

2%–6% of its plays.⁹ Sean McVay’s Rams, from the same family, did the opposite: they essentially never used 21 personnel (0.0% of plays) and built their disguise out of 11 personnel instead. Two branches of one tree solved the same personnel problem in opposite ways, and you will meet McVay’s version in the film room.

3. Motion, which makes the defense answer first

Shanahan’s offenses move players before almost every snap: a receiver running across the formation (across or “zoom” motion), a receiver sprinting toward the quarterback at the snap (jet motion), a receiver circling behind the quarterback (orbit motion). Motion and Shifts explains each one. In this system motion does three jobs at once:

- **It asks the defense a question the offense can read.** If a defender runs across the formation with the motion man, the coverage is probably man-to-man; if the safeties rotate instead, it is probably zone. The quarterback knows which before the snap.
- **It changes the run’s arithmetic.** A receiver in jet motion toward the play adds a blocker on the edge (or a threat of a jet sweep that the edge defender must respect), and motion away from the play drags a defender out of the box.
- **It hides the play.** Because the same motion is used on runs, bootlegs and dropback passes, the motion itself tells the defense nothing. This is the illusion of complexity at work: one look, many plays.

By FTN’s charting, the 49ers used motion (before or at the snap) on more plays than all but one or two teams in every season from 2022 to 2025 (the figure is in the fingerprint section below).

4. Passes designed for yards after the catch

The last ingredient is the most distinctive. Most offenses judge a pass play by where the ball is caught. Shanahan’s judges it by what happens *after* the ball is caught.

Think about where receivers are when the ball arrives on the plays you’ve seen so far. On a bootleg the tight end is crossing the field at full speed, behind linebackers who just ran the other way. On play-action the slot receiver is running a crossing route through the space the linebackers vacated when they stepped up for the run. On a jet sweep or a screen the ball carrier already has blockers in front of him. In every case the receiver catches the ball *moving, facing upfield* and with *space ahead of him*. That is a play designed to produce YAC, yards after the catch.

The roster follows the design. The 49ers’ most famous skill players of the Shanahan years were chosen less for how they get open deep than for what they do with the ball: George Kittle, a tight end who blocks like a lineman and in 2018 gained 870 of his 1,377 receiving yards after the catch; Deebo Samuel, a receiver built like a running back, who in 2021 had 1,405 receiving yards and also ran for eight touchdowns lining up in the backfield; and Christian McCaffrey, a back who catches like a receiver.¹⁰

To measure this, you need a way to separate “the play put him in space” from “he did something with it.” That is what the next statistic is for.

YAC over expected: separating the play from the player

YAC over expected, or YACOE, is the yards a receiver gains after the catch minus the yards a model *expects* him to gain from that catch, averaged per reception. Positive means more than the situation predicted.

The idea works like CPOE for quarterbacks. Raw YAC rewards receivers whose catches come in easy places: a screen caught behind the line with three blockers in front of it will produce more yards after the catch than a 20-yard out caught at the sideline, no matter who catches either. So a model first estimates the *expected* YAC of each catch from its circumstances, and YACOE is what is left:

$$\text{YACOE on one catch} = \text{actual YAC} - \text{expected YAC}$$

A worked example: a crossing route caught 6 yards downfield, in the middle of the field, on the run. The model, having seen thousands of similar catches, expects about 6 yards after the catch. The receiver breaks a tackle and gains 15. His YACOE on that catch is +9. A screen caught behind the line where the model expects 8 and the receiver gains 5 is –3. Over a season the noise averages out and the per-catch average tells you whether a team’s receivers *consistently* beat what their catches should produce.

⁹Course calculation from nflverse participation data joined to play-by-play, regular-season runs and passes with a recorded personnel grouping, 2019–2025. Personnel is backs (RB + FB) then tight ends. The participation source changes from Next Gen Stats to FTN in 2023 and the personnel strings change format, which the code handles.

¹⁰nflverse player stats (regular seasons): Kittle 2018, 88 catches, 1,377 yards, 870 after the catch; Samuel 2021, 77 catches, 1,405 receiving yards, 59 carries, 365 rushing yards, 8 rushing touchdowns. Checks with Pro Football Reference: <https://www.pro-football-reference.com/players/K/KittGe00.htm>, <https://www.pro-football-reference.com/players/S/SamuDe00.htm>.

Two models are in common use, and they disagree in an instructive way:

- **nflverse's expected YAC** (the `xyac_mean_yardage` column in the free play-by-play) is built from what the play-by-play knows: air yards, where on the field the ball was caught, down, distance and field position. It doesn't know where the defenders were. So when a play *design* creates empty grass in front of the receiver, the nflverse model doesn't see the grass, and the design shows up as YACOE.
- **Next Gen Stats** (the NFL's tracking-data service) builds its expected YAC from player tracking: the receiver's speed and direction at the catch and the positions of the defenders around him. Space the scheme created is already in its expectation, so its YACOE leans more toward what the *receiver* did with that space.

A team that ranks high on both is probably getting yards from design *and* from players. Keep that in mind for the fingerprint section, because that is exactly what the 49ers did.

Why the four are one idea

Put the four together and each one makes the others work.

Wide zone makes linebackers flow fast and sideways; that flow is what empties the crossing lanes the YAC passes run through. The fullback forces the defense to choose between a light box (good for wide zone) and slower linebackers on the field (good for the passes). Motion tells the quarterback which coverage he is facing and moves defenders before the play starts. And all of it comes from the same few formations, so the defense cannot tell, in the first second, which ingredient is the one that matters on this snap. 08-02 draws the signature plate of this idea, "one formation, four plays" (wide zone, a naked bootleg, counter and a play-action shot, identical for the first half-second); it is worth a second look before you read on.

Signature play: wide zone against the bear

Every great offense forces defenses to invent answers. The oldest answer to wide zone is to take away the double teams it depends on. Wide zone works by pairing linemen: an uncovered lineman helps his covered neighbor reach a defender (the reach-and-overtake), then climbs to a linebacker. A defense that covers *every* interior lineman leaves no free lineman to help or to climb. That front has a name: the **bear front**, a nose on the center and a 3-technique on each guard, named after the 1985 Chicago Bears' 46 defense that made it famous. Even Fronts explains how it breaks blocking schemes in general. Figure 1 shows what it does to wide zone, and how the Shanahan offense answers.

Wide zone right vs a bear front

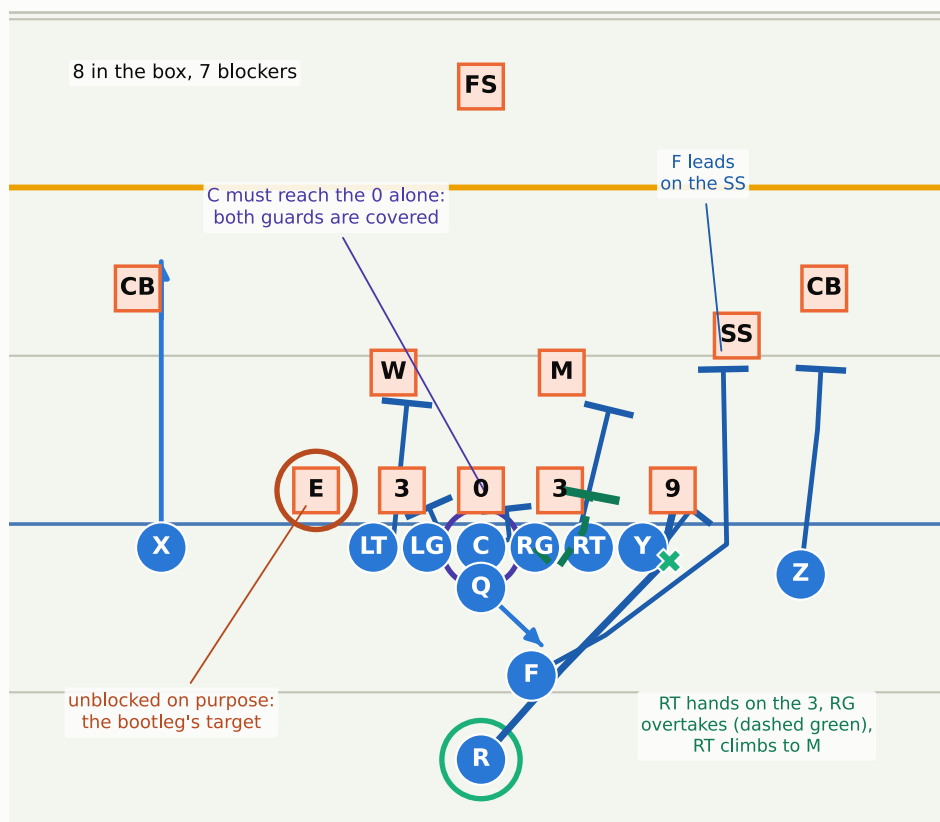


Figure 1: Wide zone right against a bear front (illustrative). The defense covers the center and both guards (a 0 and two 3s), puts a 9-technique outside the tight end and rolls the strong safety into the box: eight defenders in the box against seven blockers (five linemen, the Y and the fullback). The center (purple ring) has the hardest block in the play: he must reach a nose head-up on him with no help, because both guards are covered. The right tackle gets his hands on the play-side 3 and the right guard overtakes that block (dashed green), freeing the tackle to climb to the Mike; the Y reaches the 9; the fullback leads outside onto the safety; the left tackle climbs to the Will and the left guard cuts off the backside 3. Eight defenders, seven blockers: somebody goes unblocked, and the offense chooses who. It leaves the backside end (red ring), the defender farthest from the play, and makes him pay later with the bootleg. The back (green ring) aims at the Y's outside leg (X).

Walk through it the way a defensive coordinator would.

What the bear takes away. Look at the middle of the line. On a normal four-man front, one guard is uncovered, and that guard is the engine of zone: he helps his neighbor, then climbs to a linebacker. Against the bear, the center and both guards each have a man in front of them. The center's job is now the hardest block in football: reach a nose tackle who is lined up directly on his nose, moving the same way he is, with no one to help. If the nose wins, he is in the back's path before the back reaches the line; and because he has stepped play-side, he also clogs the cutback lane that outside zone lives on (the cutback is the back's cut against the flow, behind the defenders who over-ran the play). Meanwhile the 3-techniques sit outside the guards, the hardest alignment of all to reach. Add an eighth defender in the box and the offense has more defenders than blockers.

What the offense does about it. Three things, and you can watch for all of them.

- **Bring the fullback.** The fullback is the seventh blocker, and against an eight-man box he is the difference between one unblocked defender and two. Here he leads outside the tight end onto the safety, the man who would otherwise meet the back in the alley.
- **Choose who goes unblocked.** With eight defenders and seven blockers, someone is free. The offense picks the one who can hurt it least: the backside end, the man farthest from the ball. On a wide-zone run to the right he has to run the length of the line to make the tackle, from behind.

- **Make that choice expensive for the defense.** This is the key. The backside end has been left free *on purpose*, and an unblocked defender who is taught to chase the run is the exact defender the bootleg is built to beat. Which brings us to the play that keeps the bear honest.

! Why it exists

The bear front exists to take away the double teams and climbs that zone runs depend on, and it is a reasonable thing to show a wide-zone team on early downs. The offense's answer is not a better block; it is a better *choice*: leave the backside end unblocked, then punish him with the bootleg when he chases. That is constraint theory in one front: the defense's answer to the base play creates the conditions for the constraint play. **The trade for the defense:** the bear puts eight in the box with only one deep safety, so it is also weaker against the play-action shot down the field.

Signature play: the bootleg-keeper off wide zone

The bootleg is the quarterback faking the run one way and rolling out the other; in the **naked** version nobody blocks for him, and the keeper is the option built into it: if nobody comes to stop him, the quarterback runs. Play-Action, Bootlegs, and Screens builds the full play from a two-tight-end set. Figure 2 runs the Shanahan version: the same 21-personnel look and the same bear front as Figure 1, so you can watch what happens to the defender the run left alone.

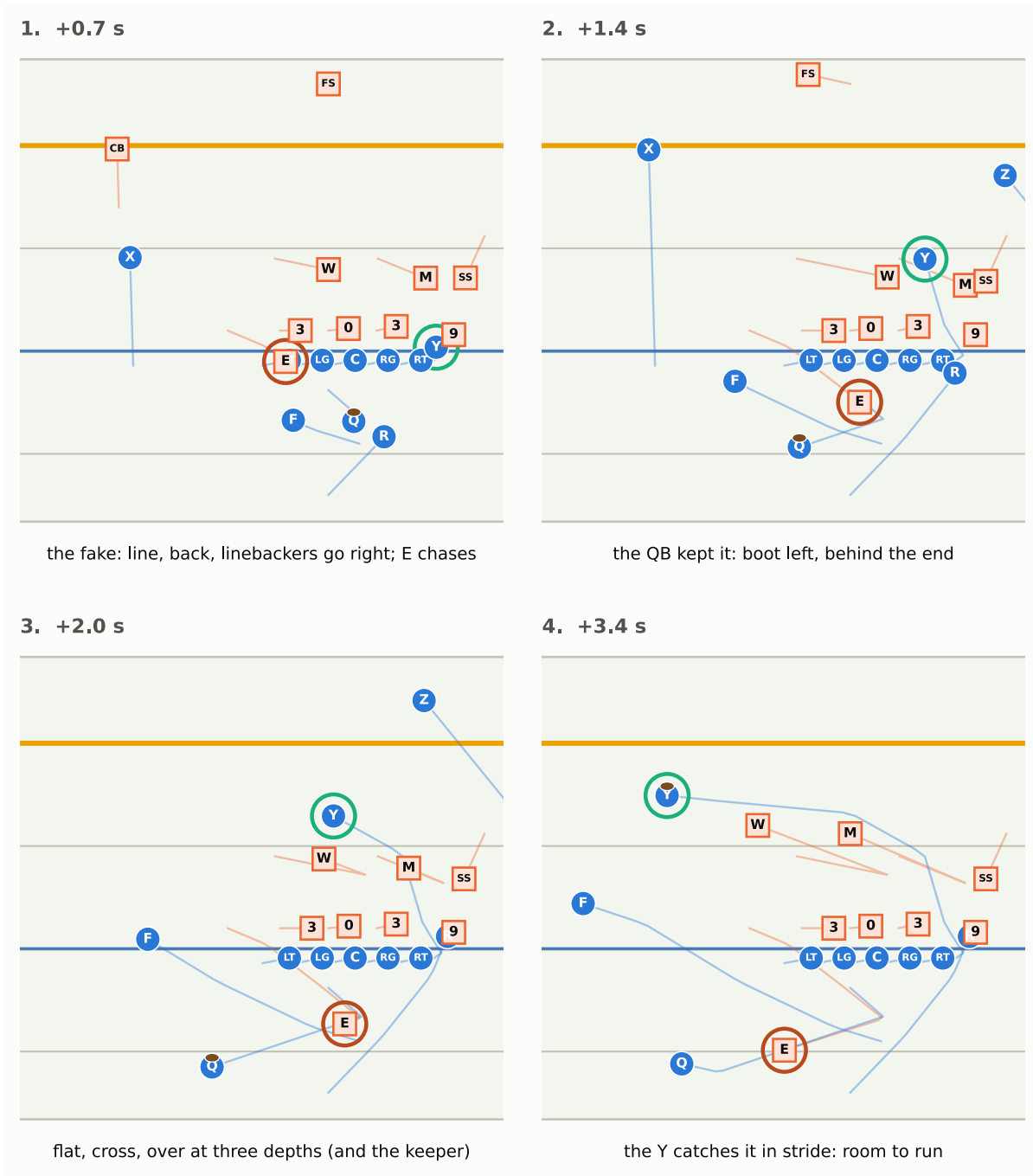


Figure 2: The bootleg-keeper off wide zone against the bear, frame by frame (simulated, BDB-format: positions in Big Data Bowl coordinates generated by gridiron; faint lines are each player's path so far). The snap is the picture in Figure 1. (1) Everything says wide zone right: the line steps right, the back runs the zone track, the linebackers and the safety flow with it, and the backside end (red ring), unblocked as on the run, chases the back down the line. (2) The quarterback never handed it off. He is rolling left behind the end; the fullback slices across toward the left flat, and the Y (green ring), who showed his run block, is crossing behind the linebackers. (3) Three receivers at three depths on the boot side: the fullback in the flat, the Y crossing at about 7 yards, the Z on a deep over route; X's deep route has carried the left corner away. Nobody is in front of the quarterback, so the keeper (running it himself) is open too. (4) The ball went to the Y; he catches it running across the field, with the Will trailing and open grass in front: yards after the catch by design.

Notice three things.

The first second is the run. Freeze the strip at frame 1 and compare it with Figure 1: the line's steps, the back's path, the quarterback's turn and the linebackers' flow are the same. A linebacker who waits to find out which play it is has already lost on the run; one who flows fast has already lost on the boot. That is why this pairing is the backbone of the system rather than a trick play.

The unblocked end is now the victim. On the run, leaving him free was cheap: he was chasing a back who was running away from him. On the boot, chasing is exactly wrong. His job against a bootleg is contain (stay outside the quarterback and force him back toward the pursuit), and he has given it up to chase a back who doesn't have the ball. The more he is taught by the run, the better the boot gets.

The read is simple and the throw is easy. The quarterback has three receivers on one side at three depths (the flat, the crosser, the deep over) and his own legs as a fourth option. He reads them high to low while moving away from the rush, and the ball he throws is short, to a receiver running into open space. That is the “easy throws” half of Kyle Shanahan's answer to the 2017 problem, and the reason this chapter's next section is about quarterbacks.

What if the end *doesn't* chase? Then he is not in the run's pursuit either, and wide zone to the other side has one fewer defender to beat. If the offense still wants to boot, it sends the fullback to block him instead of releasing to the flat: the protected bootleg of 04-05. Every defensive answer leaves something else open; the play-caller's job is to find out which answer the defense has chosen.

i Madden vs. real life

In Madden, “PA Boot” is a play you pick when you feel like it. In a Shanahan game plan, the bootleg is *earned*: it is called after the defense has been shown enough wide zone to start flowing hard, often on the very next first down after a successful run from the same look, and it is aimed at a specific defender who has been playing the run a specific way. The coaches in the booth spend the first quarter watching how the backside end and the linebackers react to the run, and the boots come later. The same call on the first snap of the game, before the defense has learned anything, is a much worse play.

The quarterback question

Here is the claim that makes the Shanahan system famous, and controversial: it makes quarterbacks efficient. Not great, necessarily; efficient. A quarterback in this offense completes a lot of passes, throws for a lot of yards per attempt, and produces strong EPA per dropback, and he does it with a smaller share of difficult throws than quarterbacks in most other systems.

Why the system helps the passer

The reasons are all in the four ingredients.

- **The run fake moves the defenders out of the throwing windows.** Linebackers who flow with wide zone are not in the middle of the field when the ball comes out on play-action; the crossing routes run behind them.
- **The bootleg halves the field and simplifies the read.** A quarterback rolling right reads three receivers on one side, high to low, with no need to scan the whole field. He also throws on the move, away from the rush, which is easier to protect.
- **Motion tells him the coverage before the snap**, so the read after the snap is a confirmation, not a discovery.
- **Receivers catch it running.** Throws are timed to receivers already moving, often across the field, so a catch tends to turn into yards. A five-yard throw that becomes a 20-yard gain counts as 20 passing yards for the quarterback.
- **Under center, the fake is better.** When the quarterback turns his back to the defense to fake a handoff, the defense can't see the ball, and the fake is more convincing than one from the shotgun (04-05 compares the two). The cost is that the quarterback has less time to read the defense on dropback passes, which the system accepts.

The debate: is that the quarterback or the system?

That design creates the argument every Shanahan-tree quarterback lives with. If the scheme manufactures open receivers and yards after the catch, how much credit does the passer deserve? It matters, because quarterback contracts are the biggest decisions a franchise makes.

The case for “system” is a list of names. Jimmy Garoppolo was efficient in San Francisco and was not after he left. Brock Purdy was the last pick of the 2022 draft, the 262nd player chosen, “Mr. Irrelevant,” and in 2023 he led the league in passer rating and yards per attempt.¹¹ Sam Darnold struggled with the Jets and Panthers, spent 2023 as Purdy's backup, then had his two best seasons in offenses from the same family: Kevin O'Connell's Minnesota Vikings in 2024 and Clint Kubiak's Seattle Seahawks in 2025, where he won Super Bowl LX. Figure 3 puts numbers on the most-cited cases.

¹¹Purdy was the 262nd and final pick of the 2022 draft. In 2023 he led qualifying passers in passer rating (113.0) and yards per attempt (9.6): <https://www.pro-football-reference.com/players/P/PurdBr00.htm>; <https://www.pro-football-reference.com/years/2023/passing.htm>; also NBC Sports Bay Area, <https://www.nbcsportsbayarea.com/nfl/san-francisco-49ers/four-49ers-led-nfl-key-stats-2023/1689409/>.

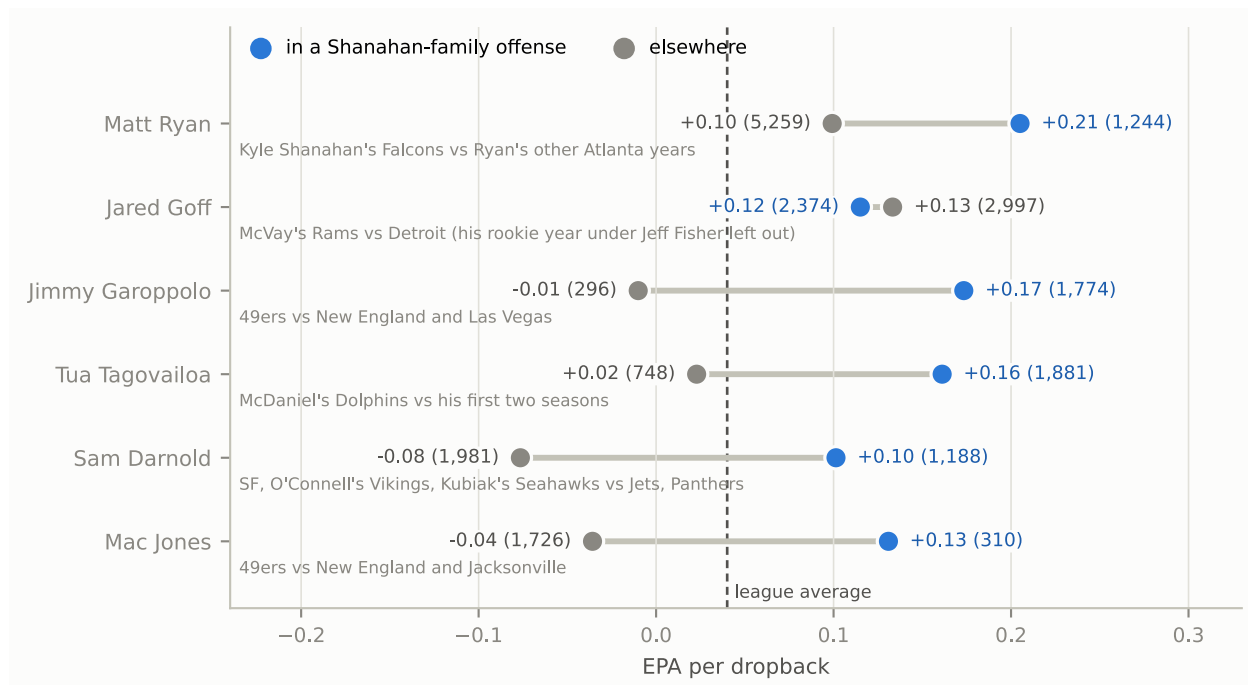


Figure 3: Same quarterback, different offense: EPA per dropback (every dropback, including sacks and scrambles; regular seasons) inside and outside a Shanahan-family offense. Blue dots: seasons with a Shanahan-family play-caller; grey dots: the same quarterback's other seasons (the number of dropbacks is beside each dot). The dashed line is the league average dropback, 2012–2025. Most of these quarterbacks were much better inside the family. The exception is the point of the debate: Jared Goff was about as good in Detroit (2021–2025) as he had been under McVay, in an offense whose coordinator from 2022 to 2024, Ben Johnson, borrowed heavily from the same ideas. Samples are small and nothing here controls for age, teammates or opponents. Data: nflverse play-by-play.

Read Figure 3 with care, because it is the kind of chart that can prove whatever you want. Darnold's jump from -0.08 to $+0.10$ EPA per dropback and Mac Jones's from -0.04 to $+0.13$ are large, and they happened with new teammates, new offensive lines and, in Darnold's case, a few more years of experience. Tagovailoa is the cleanest comparison (same team, same player, a new head coach in 2022), and his line hides a collapse: his 2025 season under McDaniels was his worst since his rookie year, and McDaniels was fired that winter. Garoppolo's drop after leaving San Francisco came at age 32 and after injuries. And Goff, traded by McVay to Detroit for Matthew Stafford in 2021, averaged $+0.13$ per dropback in Detroit against $+0.12$ in Los Angeles, and had his best season (2024) after the Lions built their own run-and-play-action offense around him.

So the honest summary is in between. The system reliably raises the *floor*: quarterbacks who can read a defense, deliver the ball on time and throw on the move become efficient in it, and some of them have looked like stars. What the system cannot fully supply is the throw outside the design: the third-and-11 where the coverage was right, the protection broke down and the quarterback has to create something. That is why McVay traded Goff for Stafford in 2021 (and won Super Bowl LVI with him the next February), and why the 2025 Rams, with the same coach and Stafford at 37, could lead the league in play-action and still have the MVP at quarterback: Stafford threw a league-leading 46 touchdown passes.¹²

Common misconception

“System quarterback” means “bad quarterback.” It doesn't, and the phrase is used carelessly in both directions. Every quarterback plays in a system; the Shanahan system just makes its contribution unusually visible. Matt Ryan's MVP season (2016), Aaron Rodgers's MVP seasons under Matt LaFleur (2020 and 2021) and Stafford's in 2025 all came in this family's offenses, from quarterbacks nobody calls system players. The useful question isn't “system or quarterback?” but “how much does this quarterback add *beyond* what the design gives him?”, which is what EPA, CPOE and YAC over expected try to separate.

¹²The Rams traded Goff and draft picks to Detroit for Stafford in January 2021 and won Super Bowl LVI, 23–20 over Cincinnati, on Feb 13, 2022: <https://www.pro-football-reference.com/boxscores/20220130ram.htm>. Stafford was the 2025 AP MVP with a league-leading 46 touchdown passes (this book's checked facts, from <https://amp.nfl.com/news/list-of-nfl-honors-award-winners-from-2025-nfl-season>).

The critics have one more argument: the big games. Kyle Shanahan's offenses have led in the second half of three Super Bowls and lost all three: as Atlanta's coordinator in Super Bowl LI (up 28–3 in the third quarter, lost 34–28 in overtime), and as the 49ers' head coach in Super Bowl LIV (up 20–10 in the fourth quarter, lost 31–20) and Super Bowl LVIII (lost 25–22 in overtime).¹³ Those losses get pinned on play-calling with a lead and on an offense that, critics say, can't throw its way out of obvious passing downs. The second claim is testable, and the data doesn't support it. From 2019 to 2025, on third down with 7 or more yards to go (the downs where everybody knows it is a pass), the 49ers' dropbacks averaged +0.18 EPA, 1st of 32 teams, against a league average of –0.10. Whatever went wrong in those Super Bowls, it wasn't that the system has no answer on third-and-long.

The data fingerprint

How would you recognise this offense from data alone, without watching a snap? Four measurements do most of the work: how often the quarterback is under center, how often he fakes a handoff, how much the receivers gain after the catch, and (to check that all this produces anything) how efficient the offense is.

Under center and play-action

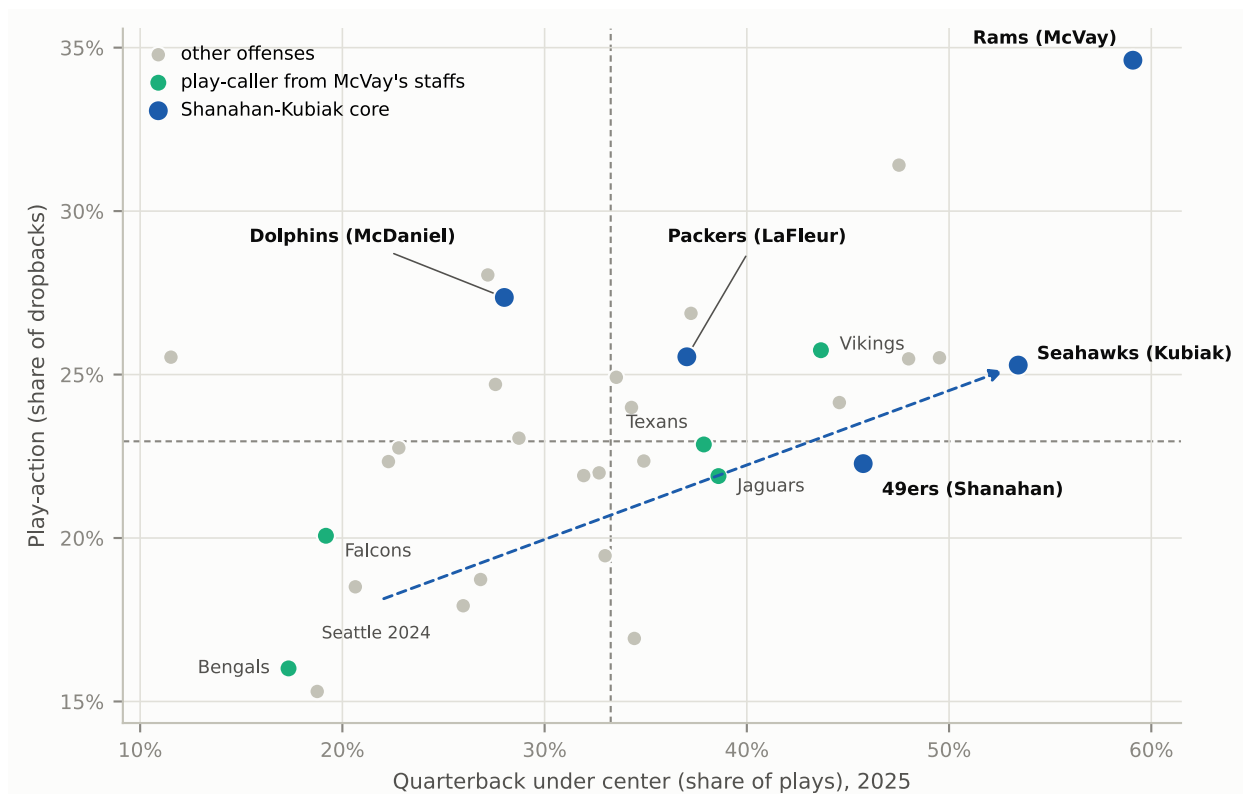


Figure 4: The 2025 fingerprint: how often each offense put the quarterback under center (share of runs and passes) against how often it used play-action (share of dropbacks), from FTN's charting. The five offenses with play-callers from the Shanahan-Kubiak core are in dark blue; five whose play-callers came up on Sean McVay's staffs are in green; the rest of the league is grey; dashed lines are the league medians. Four of the five core teams sit right of the under-center median, the Rams at the far corner of both; McDaniel's Dolphins, the speed-and-motion branch, worked mostly from the shotgun and pistol. The arrow shows Seattle's move from 2024 to 2025, Klint Kubiak's one season as play-caller. The second generation is spread out: a coaching tree passes on ideas, not a single formula. Data: nflverse play-by-play joined to FTN charting, 2024–2025 regular seasons.

The core of the family is easy to spot in Figure 4. In 2025 the Rams put Stafford under center on 59% of their plays, the most in the league, and used play-action on 35% of dropbacks, also the most. The 49ers were under center on 46% (6th). Seattle made the most dramatic move of any team in the chart: under center on 22% of plays in 2024, 25th in the league, and on 53% in 2025, 2nd, the year Klint Kubiak arrived as offensive coordinator. One new play-caller, and the fingerprint moved across the whole chart.

¹³Super Bowl LI (Feb 5, 2017): New England 34, Atlanta 28 (OT), after Atlanta led 28–3 in the third quarter: <https://www.pro-football-reference.com/boxscores/201702050atl.htm>. Super Bowl LIV (Feb 2, 2020): Kansas City 31, San Francisco 20, after the 49ers led 20–10 in the fourth quarter: <https://www.pro-football-reference.com/boxscores/202002020kan.htm>. Super Bowl LVIII (Feb 11, 2024): Kansas City 25, San Francisco 22 (OT). Scores also check against nflverse schedules.

Two surprises are worth noticing. The first is San Francisco's play-action rate, which in FTN's charting is ordinary: 22% of dropbacks in 2025, 21st in the league, and in the bottom half in every season from 2022 to 2025. Shanahan's offense gets its "it all looks the same" effect less from volume of fakes than from the run game, the motion and the boots and keepers that FTN counts with them. Where the 49ers *are* extreme is motion (2nd in 2025, 2nd, 2nd and 3rd in the three seasons before) and personnel. The second surprise is how scattered the second generation is: the Bengals under Zac Taylor are a shotgun passing team, and Atlanta in 2025 was under center less than almost anyone. A coaching tree hands down a way of thinking and a vocabulary, not a formation chart.

That second point also tells you the fingerprint has been *spreading*. League-wide, quarterbacks were under center on 27% of plays in 2023 and 34% in 2025 by FTN's charting, the reversal of a decade-long slide toward the shotgun.¹⁴ Offenses that are not in the tree have been borrowing its habits.

YAC over expected

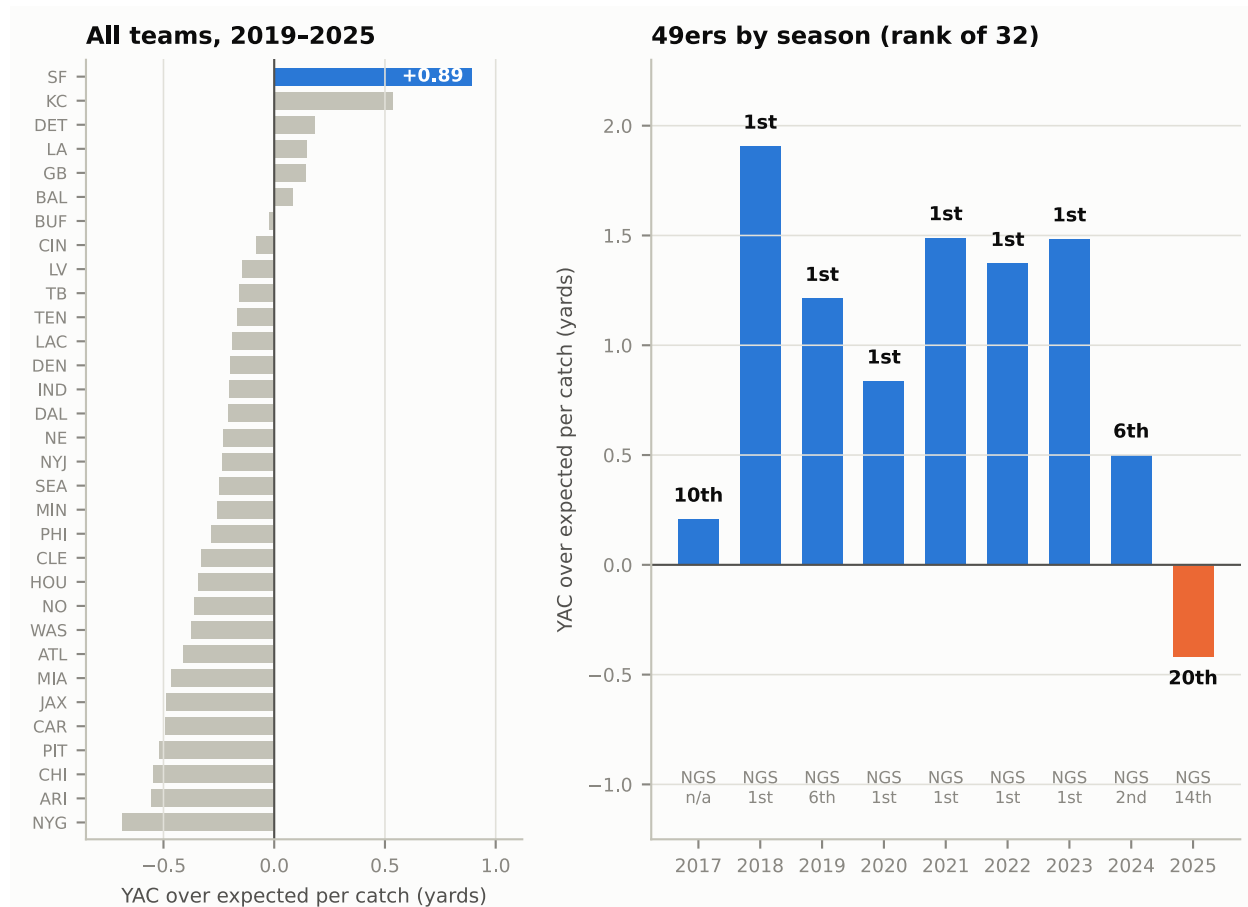


Figure 5: YAC over expected, the 49ers' signature. Left: every team's yards after the catch per completion above nflverse's expected YAC, pooled over the 2019–2025 regular seasons; San Francisco (blue) is first by a wide margin. Right: the 49ers season by season, 2017–2025, with their rank among the 32 teams above each bar (nflverse model) and below it in grey the rank by Next Gen Stats' tracking-based model (available from 2018; NGS team values are reception-weighted averages of the receivers it lists). Both models put San Francisco first in most seasons from 2018 to 2023. The 2025 drop came in a season when the receiving corps changed and Mac Jones started about half the games. Data: nflverse play-by-play; NGS receiving via nflreadpy.

No team in the period comes close to San Francisco in Figure 5. From 2019 to 2025 the 49ers' receivers gained +0.89 yards after the catch per reception beyond what nflverse's model expected, against +0.54 for the next team, KC. They ranked first in every one of the six seasons from 2018 to 2023, a stretch that began with C. J. Beathard and Nick Mullens throwing to Kittle in 2018. Among receivers with 40 or more catches from 2018 to 2025, Deebo Samuel's 2021 ranks 1st in Next Gen Stats' YAC over expected: +5.2 yards per catch above what the tracking model expected.

¹⁴Course calculation, nflverse play-by-play joined to FTN charting (`qb_location == "U"`), regular-season runs and passes. The same series is in this book's checked facts (FACTS-current §10).

The two models agreeing matters. nflverse’s model doesn’t know where the defenders were, so it credits the 49ers for space the *design* created; Next Gen Stats’ model does know, and it still ranked them first in most of those seasons, so their receivers were also doing more with the space than the space alone predicts. Design and players, together. The 2025 bars are the reminder that YAC is partly personnel: Deebo Samuel was traded to Washington before the 2025 season, and Mac Jones started about half the games.¹⁵

Does it produce points?

All of that would be trivia if the offense weren’t good. Figure 6 shows where San Francisco’s offense ranked in EPA per play in each season from 2019 to 2025, with the quarterbacks who took its dropbacks.

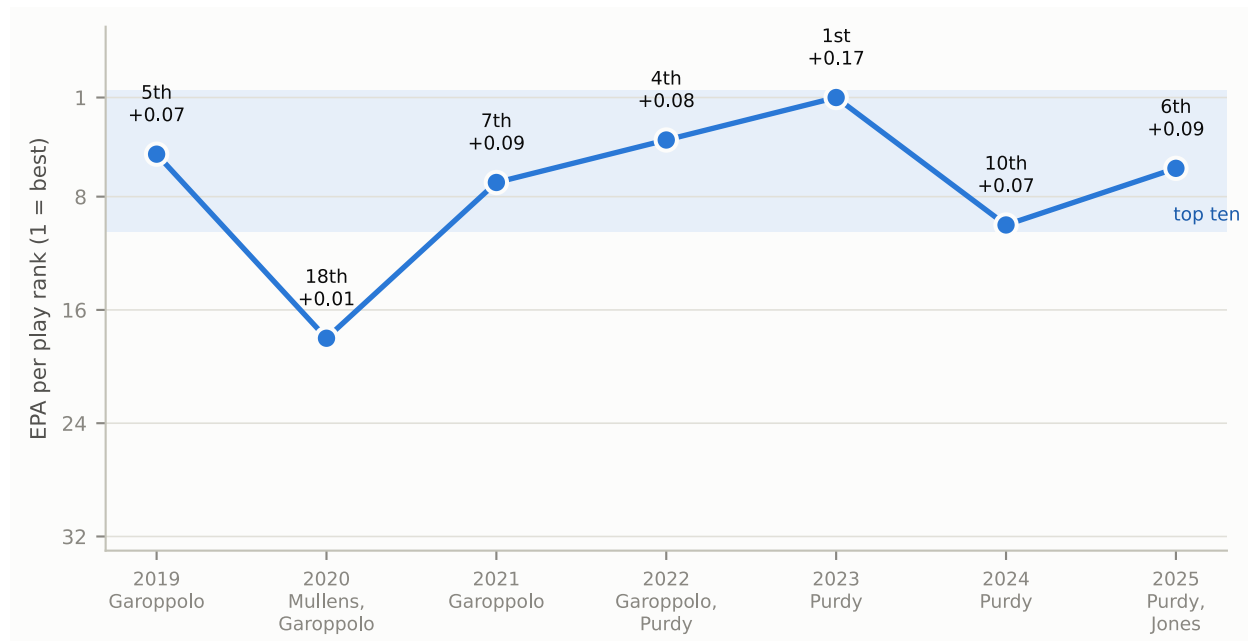


Figure 6: San Francisco’s offense, EPA per play rank among the 32 teams, 2019–2025 regular seasons (all runs and passes; 1st = most efficient), with the quarterbacks who took the most dropbacks each season beneath (a second name if he took at least a quarter of them). Six of the seven seasons are in the top ten (shaded), with the dropbacks spread across several quarterbacks. The outlier, 2020, was a season of injuries in which three quarterbacks started. Data: nflverse play-by-play.

Six of seven seasons in the top ten, including first in 2023 (Purdy’s league-leading season, with McCaffrey the Offensive Player of the Year), with three different quarterbacks leading the team in dropbacks along the way and four taking at least a quarter of them in some season. Before 2019 the system was still being built: the 49ers ranked 16th in 2017 and 24th in 2018, with Hoyer, Beathard, Garoppolo and Mullens all starting games. That steadiness across passers is the strongest single piece of evidence for the system’s claim. It isn’t proof (the 49ers also had elite players at tight end, running back, receiver and left tackle in most of those seasons), but no other offense of the period stayed this efficient through this much change at quarterback.

¹⁵Samuel’s trade to Washington for a fifth-round pick was agreed on March 1, 2025 and became official when the league year opened on March 12: ESPN, “49ers deal Deebo Samuel to Commanders, sources say,” https://www.espn.com/nfl/story/_/id/44071829/49ers-deal-deebo-samuel-commanders-sources-say. Jones and Purdy split the 49ers’ 2025 dropbacks almost evenly in nflverse play-by-play (Purdy 9 starts, Jones 8).

i Film room: Atlanta Falcons 2016

The season: Kyle Shanahan's second and last as Atlanta's offensive coordinator. The Falcons led the NFL with 540 points, Matt Ryan was named MVP, and they reached Super Bowl LI, where they led New England 28–3 in the third quarter and lost 34–28 in the first overtime in Super Bowl history.¹⁶

What to notice:

- This is the system before San Francisco: outside zone with two backs who could both run and catch (Devonta Freeman and Tevin Coleman), bootlegs off it, and play-action shots to Julio Jones.
- Watch the running backs as receivers. Much of the Shanahan YAC game in its later 49ers form (the back leaking out of the backfield, the screen against a blitz) is already here.
- Watch Ryan's feet on play-action. He turns his back to the defense under center, carries out the fake, and throws to a receiver who is already open, often crossing. It is the "easy throw" in action.

i Film room: Miami Dolphins 2023

The season: Mike McDaniel's second as Miami's head coach. McDaniel spent most of his career on Kyle Shanahan's staffs (in Houston, Washington, Cleveland, Atlanta and San Francisco, where he was run-game coordinator and then, in 2021, offensive coordinator) and brought the system to Miami in 2022 with a twist: speed. In Week 3 of 2023 the Dolphins beat Denver 70–20; Tua Tagovailoa led the league with 4,624 passing yards and Tyreek Hill had 1,799 receiving yards.¹⁷

What to notice:

- Motion on almost every snap, much of it at the snap, the most of any team in the league (by FTN's charting Miami was first in motion in 2022, 2023 and 2025). Three weeks into 2023, ESPN's tracking had Miami with a man in motion *at* the snap on 59% of plays, the highest rate for any team since ESPN began counting in 2017.¹⁸
- The fullback. Miami was the league leader in 21 personnel in 2022 and 2023 by nflverse's data, with Alec Ingold leading wide zone and slicing on boots, but Miami's backs and receivers were sprinters, so the same ideas produced longer plays.
- The limit. The Dolphins' offense faded late in 2023 and fell to the bottom half of the league in EPA per play in 2024 and 2025, and McDaniel was fired in January 2026. A system can raise a team's floor; it doesn't guarantee a ceiling, and it doesn't protect a coach.

i Film room: Los Angeles Rams 2018 and 2025

The seasons: two versions of Sean McVay's offense, seven years apart. In 2018 the Rams went 13–3 and reached Super Bowl LIII with Jared Goff, running the whole offense out of one personnel group (11 personnel on about 93% of plays, 08-02's film room). In 2025, with Matthew Stafford at 37, the Rams were first in the league in both under-center rate and play-action rate and second in EPA per play, and Stafford was MVP.

What to do: watch one game from each. In 2018 count the snaps in 11 personnel and notice how much of the deception comes from receivers' splits and motion, not substitutions; that is McVay's answer to the personnel problem the 49ers solved with a fullback. In 2025 watch the play-action passes from under center and count how often Stafford throws *before* the fake has finished, a timing only a veteran can pull off. Then compare the two quarterbacks on third-and-long: that is the debate from earlier, on film.

¹⁶Pro Football Reference, 2016 Atlanta Falcons: <https://www.pro-football-reference.com/teams/atl/2016.htm>; Super Bowl LI box score: <https://www.pro-football-reference.com/boxscores/201702050atl.htm>.

¹⁷Miami 70, Denver 20, Sept 24, 2023 (nflverse schedules); Tagovailoa 4,624 passing yards and Hill 1,799 receiving yards in 2023 (nflverse player stats), both league-leading: <https://www.pro-football-reference.com/years/2023/passing.htm>, <https://www.pro-football-reference.com/years/2023/receiving.htm>. McDaniel's career: Pro Football Reference, <https://www.pro-football-reference.com/coaches/McDaMi0.htm>; Texans offensive assistant 2006–08; Washington offensive assistant 2011–12 and receivers coach 2013; Browns receivers 2014; Falcons offensive assistant 2015–16; 49ers run-game specialist 2017, run-game coordinator 2018–20, offensive coordinator 2021 (also https://en.wikipedia.org/wiki/Mike_McDaniel).

¹⁸Seth Walder (ESPN Analytics), as reported by ProFootballTalk, "Dolphins using motion like no other NFL offense ever" (Michael David Smith, Sept 26, 2023): <https://www.nbcsports.com/nfl/profootballtalk/rumor-mill/news/dolphins-using-motion-like-no-other-nfl-offense-ever>. The 59% covers the first three weeks; later in-season reports put it higher. FTN's *is_motion* flag counts motion before *or* at the snap and is not comparable.

The tree in 2025 and 2026

A coaching tree is the family of coaches descended from one head coach, traced through who worked on whose staff. The Walsh case study draws the whole Walsh tree, of which this family is the largest modern branch: Mike Shanahan was Seifert's coordinator in San Francisco before he went to Denver. This section zooms in on the part of it that calls plays today.

The family has two roots in the Denver building of the 1990s. Gary Kubiak, Mike Shanahan's coordinator, carried the system to Houston as head coach (2006–13) and back to Denver, where he won Super Bowl 50 after the 2015 season; his sons Klint and Klay are coaches in it now. Kyle Shanahan took it through Houston, Washington, Cleveland and Atlanta to San Francisco. The decisive moment was the Washington staff of 2010–13, under Mike Shanahan with Kyle as offensive coordinator: Sean McVay coached the tight ends, Matt LaFleur the quarterbacks, and Mike McDaniels (from 2011) was an offensive assistant and then the receivers coach. All four became NFL head coaches.¹⁹ (That Washington offense also produced one of the scheme's purest seasons: in 2012 Alfred Morris, a sixth-round rookie, ran for 1,613 yards in wide zone, beside rookie quarterback Robert Griffin III and the read-option plays the staff borrowed from Baylor, which The College Laboratory covers.)²⁰

From there it branched. McVay's own staffs in Los Angeles produced head coaches Zac Taylor (Bengals, 2019), Kevin O'Connell (Vikings, 2022) and Liam Coen (Jaguars, 2025). LaFleur took the system to Green Bay in 2019; Aaron Rodgers won two MVPs in it. Kyle's 49ers staffs produced McDaniels (Dolphins, 2022), Mike LaFleur (Matt's brother) and Brian Fleury, and they gave Klint Kubiak a season on the staff before he went on to run offenses in New Orleans and Seattle.

Staffs change every winter, and this one changed a lot. Figure 7 lays out who held which job in the 2025 season and the 2026 season.

¹⁹On the 2010–13 Washington staff (Kyle Shanahan offensive coordinator, McVay tight ends, Matt LaFleur quarterbacks, McDaniels an offensive assistant): CBS Sports, "Dolphins hiring Mike McDaniels: four of NFL's youngest head coaches worked together on same staff in 2013": <https://www.cbssports.com/nfl/news/dolphins-hiring-mike-mcdaniels-four-of-nfls-youngest-head-coaches-worked-together-on-same-staff-in-2013>. Gary Kubiak: <https://www.pro-football-reference.com/coaches/KubiGa0.htm>.

²⁰Alfred Morris, 2012: 1,613 rushing yards as a sixth-round rookie: <https://www.pro-football-reference.com/players/M/MorrAl00.htm> (sixth round, 173rd overall, 2012).

Coach (how he joined the family)	2025 season	2026 season
Kyle Shanahan Mike Shanahan's son; Kubiak's Texans	49ers head coach <i>calls plays</i>	49ers head coach <i>calls plays</i>
Sean McVay Washington staff 2010–16	Rams head coach <i>calls plays</i>	Rams head coach <i>calls plays</i>
Matt LaFleur Washington, Atlanta; McVay's OC 2017	Packers head coach <i>calls plays</i>	Packers head coach <i>calls plays</i>
Mike McDaniel Kyle's staffs, Houston to SF	Dolphins head coach <i>calls plays</i>	Chargers offensive coordinator
Klint Kubiak Gary Kubiak's son; 49ers staff	Seahawks offensive coordinator <i>calls plays</i>	Raiders head coach
Mike LaFleur Kyle's staffs; Matt's brother	Rams offensive coordinator	Cardinals head coach
Brian Fleury 49ers staff 2019–25	49ers run-game coord. & TEs	Seahawks offensive coordinator <i>calls plays</i>
Klay Kubiak Gary Kubiak's son; 49ers staff	49ers offensive coordinator	49ers offensive coordinator
Kevin O'Connell McVay's OC 2020–21	Vikings head coach <i>calls plays</i>	Vikings head coach <i>calls plays</i>
Liam Coen McVay's Rams staffs	Jaguars head coach <i>calls plays</i>	Jaguars head coach <i>calls plays</i>
Zac Taylor McVay's Rams 2017–18	Bengals head coach <i>calls plays</i>	Bengals head coach <i>calls plays</i>

Figure 7: Recent staff moves: the Shanahan family’s play-callers and coordinators, 2025 season vs 2026 season (staffs as of October 2026). Each row is one coach; the left card is his 2025 job, the right card his 2026 job. Orange cards changed over the winter. ‘Calls plays’ marks the coach who called his offense’s plays that season; in 2026 it is left off where the team had not confirmed who calls them. The family lost a head coach in Miami and gained two, in Las Vegas and Arizona; Seattle replaced one family play-caller with another.

Read Figure 7 with the season labels in mind, because the two lists are easy to blur.

In the 2025 season the family’s play-callers ran the 49ers (Kyle Shanahan), the Rams (McVay, with Mike LaFleur as offensive coordinator), the Packers (Matt LaFleur), the Dolphins (McDaniel), the Seahawks (Klint Kubiak, as offensive coordinator under defensive head coach Mike Macdonald), and, a generation down, the Vikings, Jaguars and Bengals. The season ended with the family everywhere: the NFC Championship Game was Seattle against the Rams, 31–27, and Seattle went on to win Super Bowl LX 29–13 over New England.²¹

In the 2026 season the map is different. Miami fired McDaniel in January and he became the Chargers’ offensive coordinator; Klint Kubiak became the Raiders’ head coach after his one season in Seattle; Mike LaFleur left the Rams to become the Cardinals’ head coach; Seattle replaced Kubiak with Brian Fleury, the 49ers’ run-game coordinator and tight ends coach, so the Seahawks stayed in the family. In San Francisco, Shanahan still calls the plays with Klay Kubiak as offensive coordinator, and in Los Angeles Nathan Scheelhaase was promoted to replace Mike LaFleur.²²

²¹2025 NFC Championship (Jan 25, 2026): Seattle 31, Los Angeles Rams 27; Super Bowl LX (Feb 8, 2026): Seattle 29, New England 13. This book’s checked facts (FACTS-current §1–2) and nflverse schedules; https://en.wikipedia.org/wiki/Super_Bowl_LX.

²²2026 staffs from this book’s checked facts (FACTS-current §0 C2 and §9): Klint Kubiak Raiders head coach; McDaniel fired by Miami (Jan 8, 2026) and hired as Chargers offensive coordinator; Mike LaFleur Cardinals head coach; Brian Fleury (49ers staff from 2019; run-game coordinator and tight ends coach in 2025) Seattle’s offensive coordinator and, for the first time in his career, its play-caller (CBS Sports, Feb 15, 2026, <https://www.cbssports.com/nfl/news/seahawks-hire-brian-fleury-49ers-offensive-coordinator>); Scheelhaase promoted to Rams offensive coordinator; Klay Kubiak 49ers offensive coordinator, promoted in February 2025 with Shanahan still calling plays (<https://www.si.com/nfl/49ers-kyle-shanahan-promote-klay-kubiak-offensive-coordinator>). Klint Kubiak was the 49ers’ passing game specialist in 2023 before becoming Saints OC (2024) and Seahawks OC (2025). Sources include <https://www.profootballrumors.com/2026/02/2026-nfl-offensive-defensive-coordinator-search-tracker> and <https://www.ksat.com/sports/2026/02/16/seahawks-expected-to-hire-49ers-tight-end-coach-fleury-as-offensive-coordinator-ap-source-says/>. Second-generation play-callers in Figure 4 (2025): O’Connell (MIN), Taylor (CIN), Coen (JAX), Zac Robinson (ATL offensive coordinator and play-caller, from the Rams; <https://www.atlantafalcons.com/news/zac-robinson-call-plays-sideline-week-4-change>) and Nick Caley (HOU offensive coordinator and play-caller, from the Rams; <https://abc13.com/15861425>).

The lesson of the moves board is how the system *spreads*: not by teams copying film, but by assistants leaving with the playbook and the vocabulary in their heads. That is also why the family's offenses differ. A coordinator who leaves takes the ideas, but he builds them around his own quarterback, his own roster and his own instincts, which is how a shotgun team in Cincinnati and an under-center team in Los Angeles can share a grandfather.

How defenses answered

A system this successful for this long gets studied by every defensive staff in the league. Their answers fall into four groups, and you will see all four on any Sunday.

1. Change the front. The bear in Figure 1 is the oldest answer: cover the interior linemen so nobody can climb. Its cousin is the tite front (a nose and two linemen on the tackles' inside shoulders), which walls off the middle and asks the edges to win; and its opposite is a *wide* four-man front with a quick 3-technique who penetrates before the guard's reach can land (05-01 animates it). Odd and Hybrid Fronts explains why no single front stops both zone runs, so defenses change fronts by opponent and by down. The offense's counter is everything in this chapter: the fullback, the unblocked backside end and the boot, and gap runs (counter, duo) out of the same looks when the front is built for zone.

2. Keep two safeties deep and play the run with fewer men. The most important defensive trend of the 2020s, the two-high shell, is partly an answer to this family. Play-action and bootlegs work best against a defense that brings a safety down to stop the run, so defenses led by Vic Fangio and his descendants kept both safeties deep, accepted a light box, and trusted bigger fronts to hold up against the run. Two-high coverage rose from 32.9% of snaps in 2018 to 42.0% in 2025 by Next Gen Stats' count, the highest since tracking began.²³ The fight between that idea and this one is the subject of Fangio's Two-High Revolution and Macdonald's Answer, and of The Modern Era.

3. Take away the middle and make the quarterback hold the ball. The crossing routes that produce all that YAC run through the middle of the field at linebacker depth. Fill that area with defenders, rush with stunts that take a moment to develop, and the quarterback who wanted to throw to a receiver on the move is holding the ball. The most famous example is New England's plan in Super Bowl LIII, which held McVay's Rams to 3 points; The Belichick case study draws it.

4. Hunt the tells. An offense built on looking the same can be beaten if it *doesn't* quite look the same. Defensive staffs grade the running back's depth, the fullback's alignment, the receivers' splits and the direction of the motion, looking for any detail that separates the run from the boot. That is the self-scouting war of 08-02: the offense's job is to keep its tendencies balanced inside each look, and the defense's job is to find the one look that isn't.

And the system answered back. When defenses went light and two-high, the family ran the ball more and got heavier (in 2025 offenses used fewer than three wide receivers on 41.7% of plays, the highest rate in the Next Gen Stats era), and when the bear came, the boots came with it.²⁴ When defenses took away the crossers, McVay went and got a quarterback who could beat them from the pocket.

What survives

Ideas from this family are now just *how offense is played* in much of the NFL, including by teams with no connection to it:

- **Under center is back.** After a decade of shotgun growth, quarterbacks lined up under center more often in 2025 than in 2023, led by this family.
- **Motion is universal.** PFF counted a shift or motion on 63.9% of plays in 2025, up from 38.4% in 2016; every team but one used it on at least half its plays.²⁵
- **Heavier personnel is back,** for the same reason the 49ers kept a fullback: to make light defenses pay.
- **YAC as a design goal.** Receivers are now scouted and drafted for what they do after the catch, and offenses design throws to catch them on the move.
- **The illusion of complexity** (few plays, many looks; many plays, one look) is standard teaching on almost every offensive staff, inside the tree and out.

²³NFL Next Gen Stats: split-safety coverage on 32.9% of snaps in 2018 and 42.0% in 2025: <https://www.nfl.com/news/the-coverage-that-turned-the-seahawks-into-super-bowl-winners-and-is-taking-the-nfl-by-storm>.

²⁴NFL Next Gen Stats (James Reber): offenses used fewer than three wide receivers on 41.7% of plays in 2025: <https://www.nfl.com/news/nfl-trend-watch-heavy-personnel-could-shape-super-bowl-lx-showdown-between-patriots-and-seahawks>.

²⁵Daire Carragher, "NFL pre-snap motion usage reaches new highs across the NFL," PFF, Feb 23, 2026: <https://www.pff.com/news/nfl-pre-snap-motion-usage-reaches-new-highs-across-the-nfl>. PFF counts shifts as well as motion, so its series is not comparable with FTN's `is_motion` flag.

What may not survive is the specific balance. The 2025 Rams leaned harder on play-action from under center than anyone, while the 49ers' play-action rate was ordinary, and the second generation in Cincinnati and Atlanta looks more like a spread offense than like Denver in 1998. A tree spreads by mutating. That is also how you will know it is still alive: the family resemblance will be in the questions the offense asks a defense, not in any one formation.

💡 Watch for it

On Sunday: same look, different plays. Pick one Shanahan-family offense (in 2026: the 49ers, Rams, Packers, Vikings, Jaguars, Bengals, Seahawks under Fleury, the Raiders under Klint Kubiak or the Cardinals under Mike LaFleur) and keep a simple chart for one half.

Before each snap

- Write the **look**: personnel (count the backs and tight ends), where the quarterback is (under center or shotgun), and the **motion** (none, across, jet, orbit). For example, "21, under center, Z across."
- Is there a **fullback**? If so, where is he: offset to the tight end or away from it?

In the first second

- Do all five linemen step the same way? (Zone.) Does the back run toward the tight end? (Wide zone, or a fake of it.)
- Find the **backside end** and watch him. Does he chase the back, or stay home?

After the play

- Write what the play turned out to be: wide zone, gap run, bootleg, keeper, play-action shot, dropback.
- If it was a pass, where was the receiver when he caught it: **moving or standing**, and with how much space ahead? That is YAC by design.

At halftime

- For each look, list the different plays run from it. A good Shanahan-family half has several plays from each look; a look that produced only one play is a tendency the defense can find (and so can you). When a look has produced a run or two and the backside end has been chasing, start watching for the bootleg from that look.

A template:

Look (personnel, alignment, motion)	Plays seen	Backside end
21, under center, Z across	wide zone R, wide zone R, boot L	chased twice
11, gun, jet from left	jet sweep, play-action cross	stayed home

Predict the play

💡 Predict the play

Drill 1. First-and-10 from your own 35, third quarter. The offense is in 21 personnel, the quarterback under center, the fullback offset to the tight end, who is lined up on the **left** this time. The defense shows the bear again: the center and both guards covered, eight in the box, one deep safety. On the offense's last three wide-zone runs toward the tight end, the backside end (on the **right**, ringed) crashed hard down the line after the back, and one of those runs lost two yards anyway. What is the offense's best call, and which way will it go?

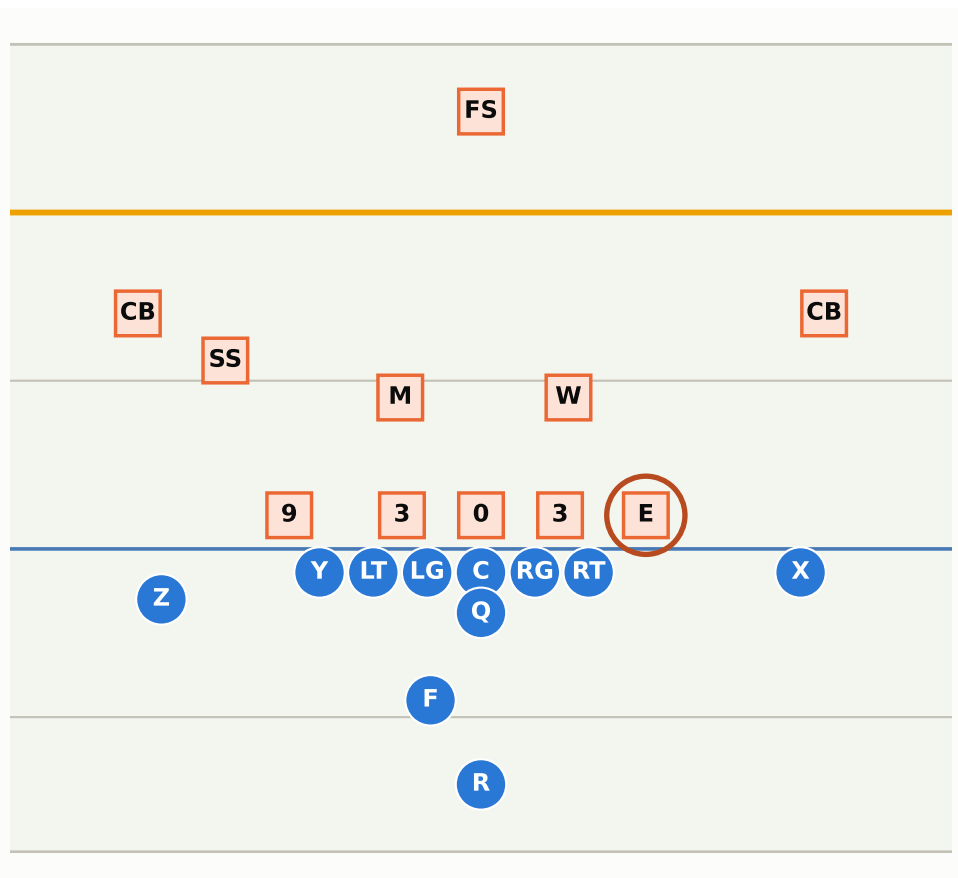


Figure 8: Drill 1: 21 personnel, offset I, the tight end on the left; a bear front with the strong safety in the box and one deep safety. The backside end (ringed) is on the offense's right.

Answer 1 is at the end of the chapter. Try it first.

💡 Predict the play

Drill 2. Second-and-6, early in the first quarter. The offense has one back and one tight end (11 personnel), the tight end in-line on the left, the quarterback under center and the receivers in tight splits. Before the snap the Z, a slot receiver on the right, goes in jet motion from right to left (dashed). The defense keeps **two** deep safeties and nobody runs across with the motion; there are six defenders in the box. Run or pass, and which way?

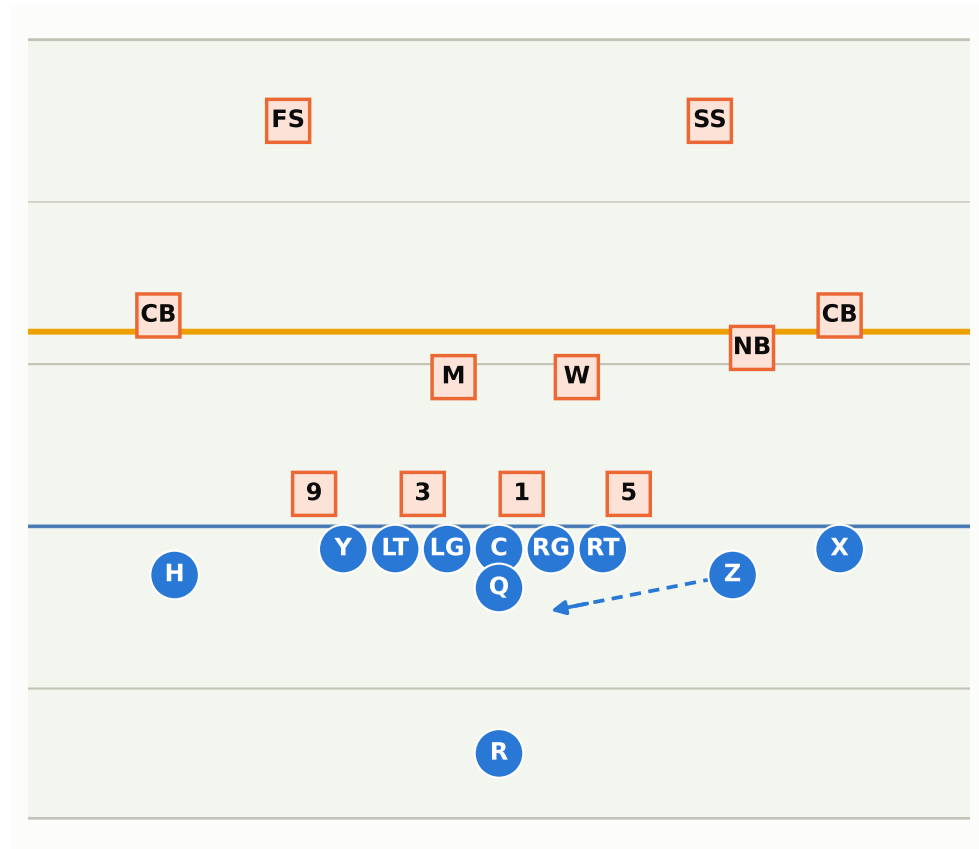


Figure 9: Drill 2: 11 personnel under center, the Y in-line on the left, receivers in tight splits; the Z, a slot receiver on the right, goes in jet motion to the left (the dashed arrow ends where he will be at the snap, just behind the quarterback). The defense: a four-man front (a 9 outside the Y, a 3, a 1 and a 5) and two linebackers in the box, a nickel back (NB) who stays outside where the Z started, two cornerbacks and two deep safeties. Nobody follows the motion.

Answer 2 is at the end of the chapter. Try it first.

💡 Predict the play

Drill 3. First-and-10, a series after the bootleg in Drill 1 went for 22 yards. Same 21-personnel look, tight end on the left, and the defense is in its bear again. But now the backside end on the right has changed: at the snap of the last wide-zone run he squeezed down the line slowly and stayed square, watching the quarterback, instead of chasing the back. The defense has stopped the boot. What should the offense call, and if it wants to boot again, what has to change?

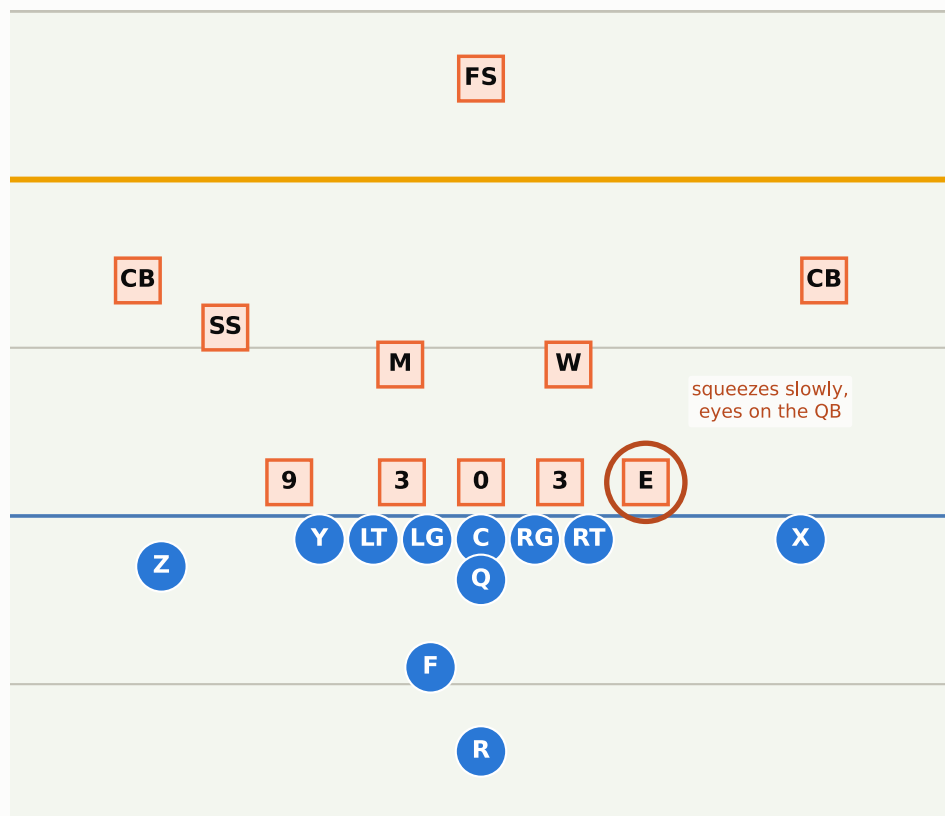


Figure 10: Drill 3: the same 21-personnel look against the same bear front. The backside end (ringed) is now playing the quarterback, not the back.

Answer 3 is at the end of the chapter. Try it first.

Takeaways

- **Two problems, one family.** Gibbs's and Mike Shanahan's 1990s Broncos built wide zone and the bootleg to block defenses that never stood still; Kyle Shanahan's 2017 49ers added the fullback, motion and YAC passing to beat fast, light defenses without a star quarterback.
- **Four ingredients, one picture.** Wide zone moves the defense sideways, the fullback forces a personnel choice, motion makes the defense answer first, and YAC passes throw into the space all of that empties. The same few looks carry all of it, which is the illusion of complexity.
- **Leave a man free on purpose, then punish him.** Against a bear front the offense can't block everyone, so it leaves the backside end and beats him with the bootleg-keeper when he chases. The defense's answer to the base play sets up the constraint play.
- **The system raises the quarterback's floor.** Several quarterbacks were much more efficient inside the family than outside it, but Goff's Detroit years and the 2021 Stafford trade show the limit: the system can't supply the throw outside the design.
- **The fingerprint:** under center and in motion, 21 or heavy personnel (49ers), play-action at the top of the league (Rams), and receivers who gain far more after the catch than the catch predicts. From 2019 to 2025 the 49ers were first in YAC over expected by a wide margin and top ten in EPA per play in six of seven seasons.

- **Label every staff by season.** In 2025 the family called plays in San Francisco, Los Angeles, Green Bay, Miami and Seattle (Super Bowl LX champions); in 2026, Kubiak and Mike LaFleur are head coaches in Las Vegas and Arizona, McDaniel is the Chargers' coordinator and Fleury calls Seattle's plays.

Connections

Builds on: Constraint Theory and Sequencing (base and constraint plays, the illusion of complexity and the “one formation, four plays” plate); The Modern Era (the spread, the two-high answer and the under-center return); Zone Running (how outside zone is blocked, the cutback and the zone-boot pairing); Play-Action, Bootlegs, and Screens (the full bootleg family); Even Fronts (the bear); Offensive Systems and Languages (the long Shanahan/McVay play call).

Sets up: Fangio's Two-High Revolution and Macdonald's Answer (the defense that grew up fighting this offense); The Lions: Ben Johnson's Offense and Dan Campbell's Aggression (a close cousin, built outside the tree); Tendencies, PROE and Charting and Tracking Data and the Big Data Bowl (build your own fingerprint and your own expected-YAC model); Building a Roster (paying for a scheme fit: fullbacks, YAC receivers and quarterbacks).

See also: Case Study: Bill Walsh's 49ers and the West Coast Offense, the tree this one grew from.

Go deeper

Chris B. Brown, *The Art of Smart Football* (2015) and his Grantland piece on Gary Kubiak's Houston run game (2012) are the best short introductions to the Gibbs zone system. Alex Gibbs's own clinic video, “Outside Zone, Tight Zone, Boots and Thoughts on the Power,” is still sold to coaches. Robert Mays's film writing and podcasts at The Ringer cover the modern Shanahan and McVay versions in detail. For the data, nflverse's play-by-play carries the expected-YAC columns used here, and Next Gen Stats publishes receiver YAC over expected on its website; Tracking Data and the Big Data Bowl shows how to build an expected-YAC model of your own from tracking data.

Answers to Predict the play

Answer 1.

A bootleg to the right (the side away from the tight end), probably naked, with the keeper built in. Wide zone here goes left, toward the tight end, so the play's backside is the offense's right, where the end has been crashing. Three runs have taught him to chase, and he is the defender the offense leaves unblocked against the bear. The quarterback fakes wide zone left, keeps the ball and rolls right behind him: the fullback slices into the right flat, the tight end crosses from left to right behind the linebackers who flowed with the fake, and if nobody comes up to stop him, the quarterback runs. First-and-10 is the ideal down for it: the defense expects run, and a short completion keeps the offense on schedule.

Why not run wide zone again? Against the bear it was losing yards; the defense has the numbers. The offense's answer is not to block the bear better but to punish the habit the bear's backside end has fallen into.

Answer 2.

A run: wide zone to the left, toward the tight end and the motion (or the jet sweep itself, handed to the Z). Count the box: six defenders against six blockers (five linemen and the tight end), with both safeties 12 yards deep. That is a light box, and the system's standing answer to two-high safeties is to run the ball at them. The motion told the quarterback two more things: nobody ran across with the Z, so the coverage is very likely zone, and the safeties didn't rotate down, so the box is staying light. Running toward the motion puts the Z's block (or the threat of the jet sweep, which freezes the edge defender) on the play side.

If the defense had answered the motion by walking a safety into the box, the picture would flip: seven in the box and one deep safety is the look the play-action shot and the boot are built for.

Answer 3.

Wide zone left again, and if they boot, a protected boot with the fullback assigned to the end. An end who stays home to wait for the quarterback is an end who isn't chasing the run. The offense left him unblocked so that he would have to chase the back from behind; now he has taken himself out of the pursuit altogether, so wide zone is effectively a play with one fewer defender against it than before. The constraint (the boot) has done its job: it changed the defense's behaviour, and the base play gets better.

If the offense wants the boot anyway, it can't be naked. The fullback, instead of slicing to the flat, slices across and *blocks* the end, giving the quarterback time to throw (the protected boot of 04-05). The cost is one fewer receiver in the flat. This back-and-forth, base play, constraint, base play, is sequencing, and it is how a Shanahan play-caller spends a game.